

MMCDs: A Multi-Stage and Multi-Model Hybrid Screening Solution for Male Contraception

Kun Li^{1,†}, Lin Miao^{2,†}, Yida Xiong^{1,†}, Yating Xu⁴, Junheng Zhu², Heyong Hu², Lei Xu², Xiangyu Wang^{5,*}, Yi Liu^{3,*}, Mengcheng Luo^{2,*} and Wenbin Hu^{1,*}

¹School of Computer Science, Wuhan University, Wuhan, 430037, Hubei, China

²State Key Laboratory of Metabolism and Regulation in Complex Organisms, Hubei Provincial Key Laboratory of Developmentally Originated Disease, TaiKang Center for Life and Medical Sciences, Taikang Medical School (School of Basic Medical Sciences), Wuhan University, Wuhan, 430071, China

³Department of Obstetrics and Gynecology, Wuhan Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, 430022, Hubei, China

⁴National Clinical Research Center for Infectious Diseases, Guangdong Provincial Clinical Research Center for Tuberculosis, Shenzhen Third People's Hospital, Southern University of Science and Technology, Shenzhen, 518112, Guangdong, China

⁵Maternal and Child Health Hospital of Hubei Province, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, 430070, Hubei, China

[†]These authors contributed equally to this work.

*Corresponding author. hwb@whu.edu.cn

Abstract

The development of male contraceptives is a critical global priority in reproductive health, essential for advancing reproductive health systems and optimizing family planning decisions. However, current research on its molecular targets is limited by target complexity and the low efficiency of compound screening, which hampers clinical translation. This study proposes a multi-stage, multi-model hybrid screening solution (MMCDs) for male contraception. MMCDs first employs two attention-based affinity classification models for large-scale compound filtering. Subsequently, a contrastive learning-based affinity regression model is integrated to quantitatively evaluate the screened compounds. Systematic protein-binding assays and cellular functional tests were conducted on the top-ranked compounds. 4 lead molecules with strong clinical translational potential were successfully identified, with the best equilibrium dissociation constant reaching 10^{-7} M. Additionally, compared to existing state-of-the-art deep learning methods, the results indicate that our solution achieves a 3.80-10.23% improvement in simulated screening performance.

Keywords male contraception, drug-target prediction, MEIOB, SPATA22, drug discovery

Introduction

Over the past few years, there have been 121 million unintended pregnancies worldwide annually, with 61% of affected women opting for abortion. Consequently, there is an urgent need for more accessible, convenient, and user-friendly contraceptives, either in the form of drugs or devices (Bearak et al. 2020). However, no effective drugs for males have been established to date. Hormonal male contraceptives, such as the male-targeted formulations dimethandrolone undecanoate (DMAU) and NES/T (progesterone + testosterone gel), have demonstrated promising contraceptive potential in clinical trials (Ilani et al. 2012; Sharma et al. 2022). Nevertheless, hormonal contraceptives are associated with side effects such as decreased libido and mood swings. Thus, in recent years, the development of non-hormonal male contraceptives has attracted extensive attention. Examples include the retinoic acid receptor α inhibitor YCT-529 and the histone deacetylase inhibitor MS-275, which have exhibited favorable contraceptive effects (Hong et al. 2024; Mannowetz et al. 2025). However, the targets of these

agents are present in multiple tissues throughout the body and are not specific to the reproductive system.

MEIOB is a meiosis-specific protein involved in meiotic recombination and chromosomal synapsis, predominantly expressed in the testes (Luo et al. 2013; Souquet et al. 2013). MEIOB must form a heterodimer with its partner protein SPATA22 to bind to the RPA complex and promote DNA damage repair (Ribeiro et al. 2021; Xu et al. 2017, 2022, 2025). Male mice deficient in MEIOB or SPATA22 are infertile, and mutations in these two proteins lead to oligospermia or azoospermia in human males (Gershoni et al. 2017; Guo et al. 2020; La Salle et al. 2012; Wu et al. 2021; Yao et al. 2022; Zhu et al. 2024a). Furthermore, MEIOB has been identified as a cancer-testis gene that is highly expressed in triple-negative breast cancer and melanoma. Inhibition of MEIOB expression significantly impairs the proliferative and migratory abilities of tumor cells (Gu et al. 2021; Saad and Hamed 2024; Shiohama et al. 2014). Therefore, small-molecule inhibitors targeting the MEIOB-SPATA22 complex could be developed for male contraceptives, and MEIOB is expected to serve as a novel therapeutic target for cancer.

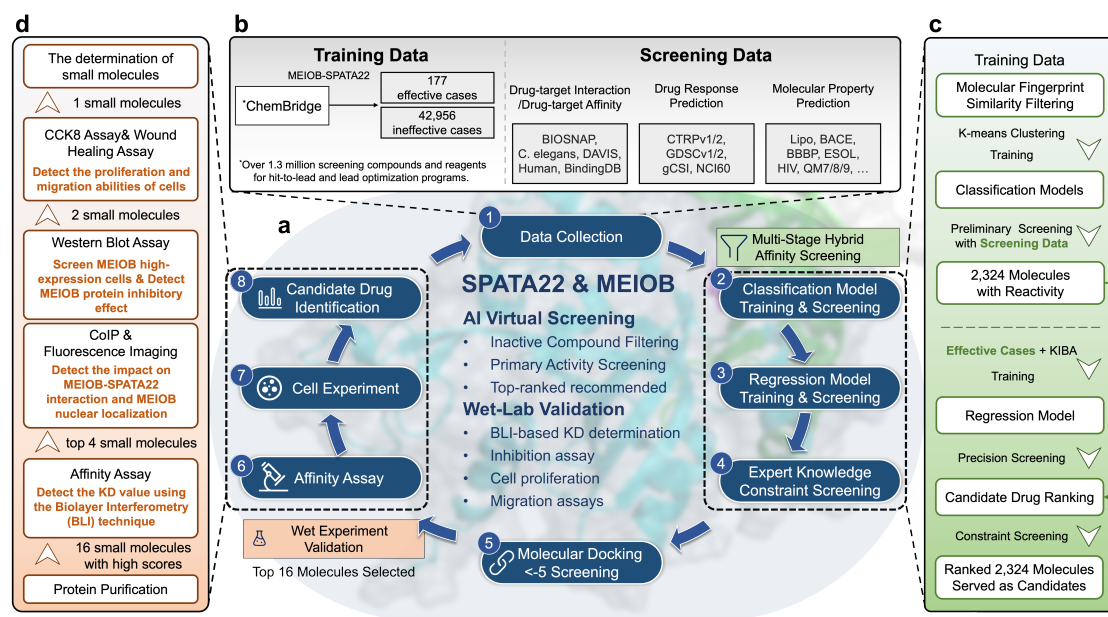


Figure 1 Multi-stage hybrid screening solution for male contraceptives. **a** Target-based multi-stage hierarchical screening solution for male contraceptive. The process includes target identification and validation, compound library preparation, virtual screening, and in vitro screening and evaluation at the cellular and molecular levels. **b** Preparation of the training and screening data. **c** The multi-stage hierarchical virtual screening workflow. **d** In vivo evaluation, including protein purification, affinity assay, Co-Immunoprecipitation (Co-IP), cell fluorescence imaging, Western blot (WB), and cell function assays.

With the rapid development of artificial intelligence (AI) technologies, the discovery of small-molecule inhibitors targeting the MEIOB–SPATA22 complex is expected to overcome the limitations of traditional high-throughput screening (Hevener et al. 2018). Compared to experimental screening, AI-driven drug discovery methods can explore a larger chemical space in a shorter time, prioritize compounds with drug-like potential, and significantly increase virtual screening hit rates, thereby minimizing the cost and time required for experimental validation (Li et al. 2025b). In recent years, studies have employed machine learning and deep learning techniques to model drug-target affinities, demonstrating promising results in virtual screening and experimental validation (Moon et al. 2022; Tian et al. 2024; Yazdani-Jahromi et al. 2022; Ye et al. 2021; Zhu et al. 2024b). On this basis, the integration of deep learning architectures such as Transformer (Vaswani et al. 2017), graph neural networks (GNN) (Scarselli et al. 2008), and convolutional neural networks (CNN) (Li et al. 2021) has significantly enhanced the multiscale feature representation of molecules and proteins. These models extract sequence information, structural patterns, as well as chemical environmental cues through end-to-end learning, enabling more efficient virtual screening and more precise activity assessment across large chemical libraries (Liu et al. 2024; Noor et al. 2024; Pandey et al. 2022; Ye and Sun 2024).

Nevertheless, applying AI methods to drug screening of the MEIOB–SPATA22 complex still faces several practical challenges (Li et al. 2024a, 2025a). The generalization ability of the models, depth of feature representation, and the complexity of the data impose significant constraints (Tiwari et al. 2023), and their performance is highly dependent on high-quality, balanced, and diverse training data (Yang et al. 2024, 2020). However, real-world biomedical data often suffer from incompleteness, bias, and noise, which severely

undermine the accuracy and stability of model predictions (Feng et al. 2024). More critically, models that perform well on standardized benchmark datasets often fail to deliver satisfactory results in real-world compound screening, with a noticeable gap between theory and practice (Mavroggiorgos et al. 2024). In compound libraries containing millions of molecules, the proportion of active compounds is typically less than 0.1%, and in extreme cases, it can be as low as 0.0001% (Wallach et al. 2024; Zang et al. 2023). This extreme class imbalance makes it difficult for models to effectively identify potential active molecules. Furthermore, proteins such as MEIOB, which are specific to the reproductive system, often have limited structural information and ligand-binding data in public databases, resulting in a lack of sufficient training samples for supervised learning models and presenting challenges related to distribution shift (Li et al. 2024b).

Therefore, this paper proposes a multi-stage, multi-model hybrid compound screening solution, termed **MMCDS**, targeting the key male contraceptive proteins MEIOB-SPATA22 (Xu et al. 2017; Wu et al. 2021). MMCDS integrates AI-driven molecule–protein affinity screening with wet-lab validation (Fig. 1). The molecular affinity screening comprises a two-stage large-scale molecular library screening followed by an affinity ranking stage using a regression-based model. The wet-lab evaluation consists of three sequential stages: affinity assay, cell experiment, and candidate drug identification. Based on the recommended results of the MMCDS solution, we performed systematic protein binding assays and cellular functional tests on the 16 molecules with the highest prediction scores, and successfully identified four lead compounds with significant clinical translational value. Their optimal equilibrium dissociation constant (KD) reaches 10^{-7} M. Pyrogallol, the most potent small molecule identified, is hypothesized to exert inhibitory effects on MEIOB function through dual mechanisms: impeding its

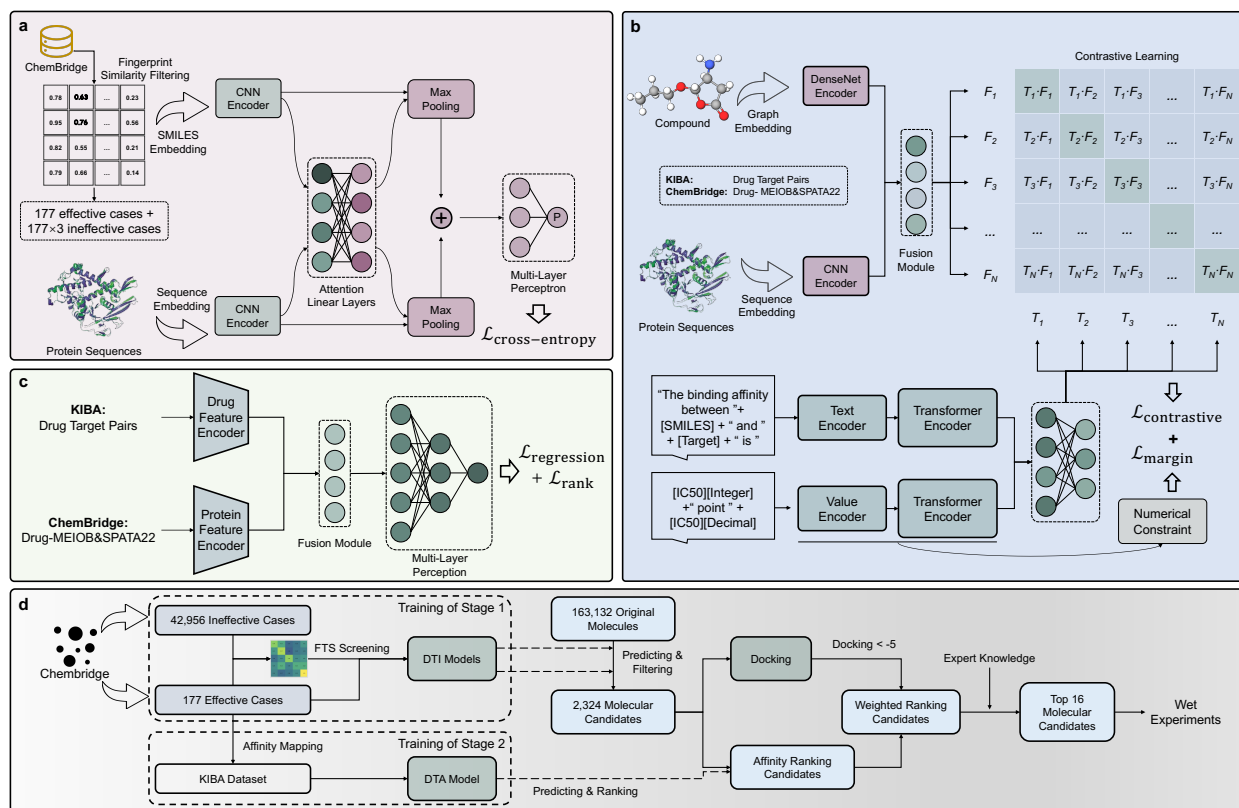


Figure 2 The model architectures used in each stage of MMCDS for the DTI/DTA tasks. **a** Classification Stage selects ineffective cases closest to the effective ones according to fingerprint similarity filtering (FTS). Then it takes SMILES strings of drugs and sequences of MEIOB & SPATA22 as input to predict their binding conditions. **b** The pre-training of the regression stage obtains the embedding of drugs' SMILES strings and protein sequences, as well as the textual description features of their interaction descriptions, aiming to predict the correct sample pairs through contrastive learning. **c** The fine-tuning of the regression stage uses the drug feature encoder and protein feature encoder in the pre-training stage. After fusing their features, the final multi-layer perceptron outputs their binding affinity. **d** Pipeline of utilizing our two models to predict and filter molecular candidates. Integrating with Schrödinger rigid docking and expert knowledge, 300 ultimate candidates were selected from 163,132 original molecules.

nuclear localization and reducing its expression levels. Furthermore, pyrogallol demonstrated the capacity to attenuate the proliferative and migratory potentials of cancer cells. Additionally, we collated and evaluated recent deep learning models under identical experimental conditions. Extensive virtual experiments validated the superiority of MMCDS over state-of-the-art (SOTA) methods for compound screening. Our solution improved MSE, PCC, and SRCC on Davis by 4.8%, 16.3%, and 9.6%, and on KIBA by 1.9%, 7.3%, and 2.2%, demonstrating strong OOD generalization.

Materials and methods

Datasets and Metrics

We use the 177 effective and 42,956 ineffective compounds derived from a previously reported cell-based high-content screen for the MEIOB-SPATA22 interaction (Xu et al. 2020) to train and test our drug-target interaction (DTI) models. We also conducted out drug-target affinity (DTA) experiments on Davis (Davis et al. 2011) and KIBA (He et al. 2017) datasets, which are the most widely used datasets in DTA research. The initial pool of candidate molecules is composed of widely used datasets from DTI, DTA, drug response prediction (DRP) and molecular property prediction (MPP), as

these molecules are synthetic, non-toxic, and potentially suitable for pharmaceutical applications. We further perform virtual molecular docking using Schrödinger on five potential pockets of MEIOB-SPATA22 to assist in evaluating the binding affinity of candidate molecules to the target.

For DTI models, we use accuracy score to evaluate the overall classification performance and recall score to measure the ability to recover true effective pairs. For DTA models, we use mean squared error (MSE) to measure the deviation between predicted and true binding affinities, and we further use Pearson correlation coefficient (PCC) and Spearman rank correlation coefficient (SRCC) to assess the linear correlation and rank consistency between predictions and ground-truth values, and Rank score (Liu et al. 2019) to evaluate the model's ability to prioritize high-affinity candidates.

MMCDS's framework

To address the key challenges that impair real-world AI-driven drug screening, such as poor data quality, extreme class imbalance, and scarce information on male contraceptive targets, we have designed a multi-stage, multi-model hybrid screening solution named MMCDS. This solution is explicitly structured into two core computational stages to systematically evaluate and prioritize potential candidates

for non-hormonal male contraceptives (Fig. 1). The first stage focuses on screening potential compounds in large-scale data. Since a single classification model cannot simultaneously filter out inactive molecules and extract active ones accurately, we apply two different training strategies to the same DTI model. These strategies effectively enable large-scale elimination of inactive molecules and maximize the recall of potential active molecules. The second stage then focuses on affinity prediction and ranking, where a DTA model is utilized to perform precise binding affinity predictions for the filtered candidates. These molecules are then ranked based on their predicted affinity to efficiently identify high-potential lead compounds.

Hierarchical Screening with Hard-negative Refinement

We adopt a coarse-to-fine screening strategy to address severe class imbalance and the discrepancy between binding signals and true inhibitory activity. A single-stage classifier is easily dominated by abundant inactive molecules and therefore tends to learn only a coarse decision boundary.

In the coarse screening stage, molecules with strong binding affinity are treated as positive samples, whereas molecules with weak affinity are treated as negative samples. To mitigate the severe class imbalance, the negative set is constructed from the top- k molecules with the highest fingerprint similarity to the positive samples. This strategy keeps the training set compact while retaining the most informative negatives, helping the model capture subtle structure-activity differences and achieve high-recall preliminary filtering. In the fine screening stage, we further separate likely active compounds from weakly binding-associated samples using a threshold of -4.2 derived from the empirical distribution of binding values. These ambiguous samples, together with the top- k negatives, form a more challenging dataset for fine-grained discrimination, thereby refining the decision boundary toward true active compounds.

The model takes the SMILES sequence of a drug S_d and the amino acid sequence of a target protein S_p as input. They are encoded into latent representations $\mathbf{H}_d$ and $\mathbf{H}_p$ through embedding layers. The two representations are then integrated by a fusion module to obtain a joint feature vector $\mathbf{h}$, which is finally fed into a multi-layer perceptron to produce the prediction score $\hat{y}$:

$$\begin{aligned} \mathbf{H}_d &= \text{Embedding}_d(S_d), \quad \mathbf{H}_p = \text{Embedding}_p(S_p), \\ \mathbf{h} &= \mathcal{F}_{\text{fusion}}(\mathbf{H}_d, \mathbf{H}_p), \quad \hat{y} = \text{MLP}(\mathbf{h}). \end{aligned} \quad (1)$$

Let y denote the ground truth labels and $\hat{y}$ the predicted outputs. The model is trained via cross-entropy loss function, defined as:

$$L(y, \hat{y}) = -\frac{\sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]}{N}. \quad (2)$$

Text-guided Affinity Ranking with Contrastive Learning

After the coarse-to-fine screening stage, the remaining candidates are further prioritized by predicted binding affinity, and the top N molecules are selected as the final recommendations. Since the MEIOB-SPATA22 complex involves only two target sequences, directly training an affinity predictor on the target-specific data may lead to limited representation diversity and weak generalization. To address this issue, we adopt a pre-training and fine-tuning strategy for affinity ranking. In the pre-training stage, drug-protein structural representations are aligned with natural language descriptions of their interactions through contrastive learning, enabling the model

to capture transferable semantics beyond the narrow target space. In the fine-tuning stage, the structural encoder is optimized for affinity prediction, and a ranking objective is further introduced to improve the relative ordering of candidate molecules.

We design a natural language-supervised contrastive learning framework during the pre-training phase (Fig. 2b). Specifically, for each input drug and protein pair (drug, protein), the drug molecule is encoded using a multi-layer graph convolutional network (GCN) with dense connections to enhance atomic-level information flow, thereby capturing both local and global features. Their features are fused into a joint representation $\mathbf{F}$:

$$\mathbf{F} = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1[\mathbf{H}_d; \mathbf{H}_p]), \quad (3)$$

where $\mathbf{H}_d$ and $\mathbf{H}_p$ denote the encoded drug and protein features, respectively, and $[\cdot; \cdot]$ denotes feature concatenation.

To provide semantic supervision beyond the narrow target space, the natural language description associated with each drug-protein pair is encoded by a Transformer encoder to obtain a textual representation $\mathbf{T}$. We then use a contrastive objective to align matched structure-text pairs in the shared embedding space. Let

$$s_{ij} = \text{sim}(\mathbf{F}_i, \mathbf{T}_j) \quad (4)$$

denote the similarity between the i -th structural representation and the j -th textual representation. The contrastive loss is defined as

$$\mathcal{L}_{\text{con}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^N \exp(s_{ij}/\tau)}, \quad (5)$$

where τ is the temperature parameter and N is the batch size.

Besides semantic alignment, we further consider that textual descriptions often contain numerical information related to interaction strength or ordered relations, which cannot be fully captured by standard text encoding alone. To preserve such ordinal information, we transform the numerical text features into structured relational representations and introduce a margin-based objective inspired by knowledge graph modeling. Specifically, for each triplet $(h, l, t) \in \mathcal{S}$, where $h, t \in \mathcal{E}$ denote entities and $l \in \mathcal{L}$ denotes a relation, the margin loss is defined as

$$\mathcal{L}_{\text{margin}} = \sum_{(h, l, t) \in \mathcal{S}} [\gamma + d(\mathbf{h} + \mathbf{l}, \mathbf{t}) - d(\mathbf{t} + \mathbf{l}, \mathbf{h})]_+, \quad (6)$$

where $[x]_+ = \max(0, x)$, $\gamma > 0$ is the margin parameter, and $d(\cdot, \cdot)$ denotes the distance function. In this way, numerical order information in the text branch is explicitly regularized and can complement the semantic alignment learned by contrastive pre-training.

The overall pre-training objective is therefore written as

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\text{con}} + \mathcal{L}_{\text{margin}}. \quad (7)$$

In the fine-tuning stage, the pre-trained structural branch is retained for affinity prediction. The fused representation $\mathbf{F}$ is fed into an MLP regressor to predict the binding affinity $\hat{y} = \text{MLP}(\mathbf{F})$. To improve both numerical accuracy and relative ranking quality, we jointly optimize the model with a mean squared error loss and a

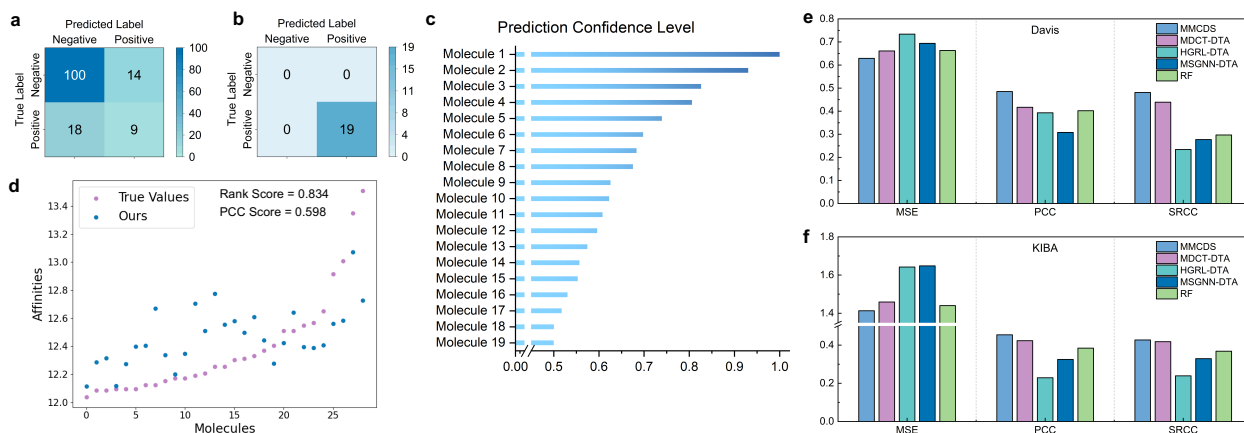


Figure 3 MMCDS's test results during the classification screening and regression ranking stages. **a,b** Confusion matrices for the two training strategies in the initial classification stage: **a** focuses on filtering out a large number of inactive molecules, while **b** aims to retain potential active molecules. **c** Prediction confidence level of 19 test molecules in the first stage. **d** Comparison of predicted affinities by DTA model and true labels in the second stage. **e, f** OOD performance comparison of the DTA model against representative baseline methods on the Davis and KIBA datasets.

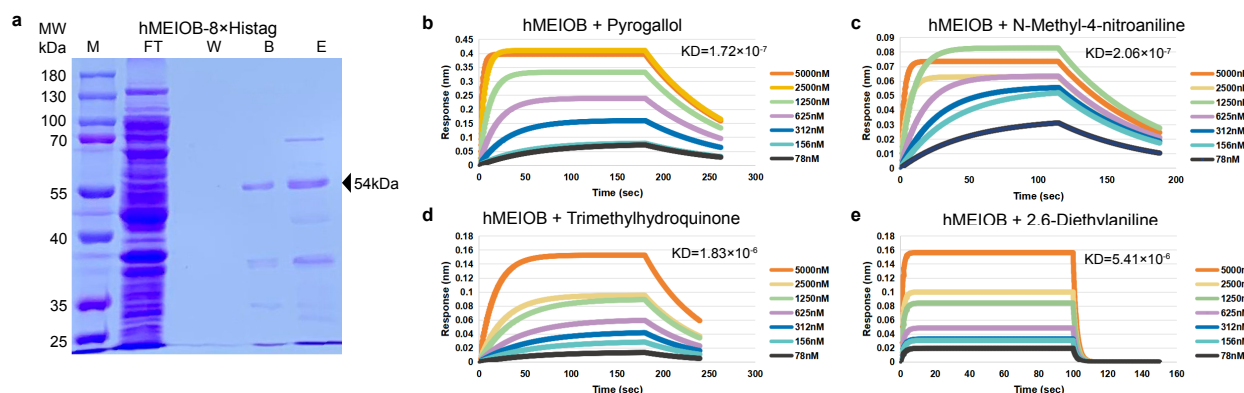


Figure 4 Affinity detection of small molecules for MEIOB. **a** Purify the full-length human MEIOB-8 × His-Tag protein from *E. coli* using Ni-NTA beads. M: Markers; FT: Flow-through; W: Wash; b: Beads; E: Elution. **b** Fitting curve of Pyrogallol and MEIOB. **c** Fitting curve of N-Methyl-4-nitroaniline and MEIOB. **d** Fitting curve of trimethylhydroquinone and MEIOB. **e** Fitting curve of 2,6-Diethylaniline and MEIOB.

ranking loss:

$$\mathcal{L}_{\text{mse}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2, \quad (8)$$

$$\mathcal{L}_{\text{rank}} = \sum_{i < j} \max(0, -\text{sign}(y_i - y_j)(\hat{y}_i - \hat{y}_j)), \quad (9)$$

The final fine-tuning objective is

$$\mathcal{L}_{\text{fine}} = \mathcal{L}_{\text{mse}} + \lambda \mathcal{L}_{\text{rank}}. \quad (10)$$

where λ controls the contribution of the ranking term.

Results

To systematically evaluate MMCDS, we organized the experimental study into two sequential parts: computational evaluation and biological validation. The computational part was designed to assess whether the proposed two-stage framework could accurately filter large-scale compound libraries and prioritize candidates with high binding potential toward the MEIOB-SPATA22 complex. The

biological part was designed to verify whether the selected compounds truly bind MEIOB and whether they further affect the MEIOB-SPATA22 interaction and its downstream cellular phenotypes.

Computational Evaluation

This section evaluates the effectiveness and state-of-the-art nature of the MMCDS method from the perspective of virtual screening. This part of the study is primarily divided into three levels. First, we evaluate the performance of the hierarchical screening stage, focusing on its ability to retain potential active compounds while removing invalid molecules. Second, we evaluate the performance of the affinity ranking stage, focusing on its generalization ability under out-of-distribution settings. Finally, we apply the complete computational workflow to screen large-scale molecular libraries and use the resulting candidate molecules for downstream wet-lab validation.

Large-scale high-precision compound filtering

In the first stage, the primary objective is to filter out molecules that do not exhibit significant interactions with the MEIOB-SPATA22 complex, a protein complex involved in the regulation of male fertility,

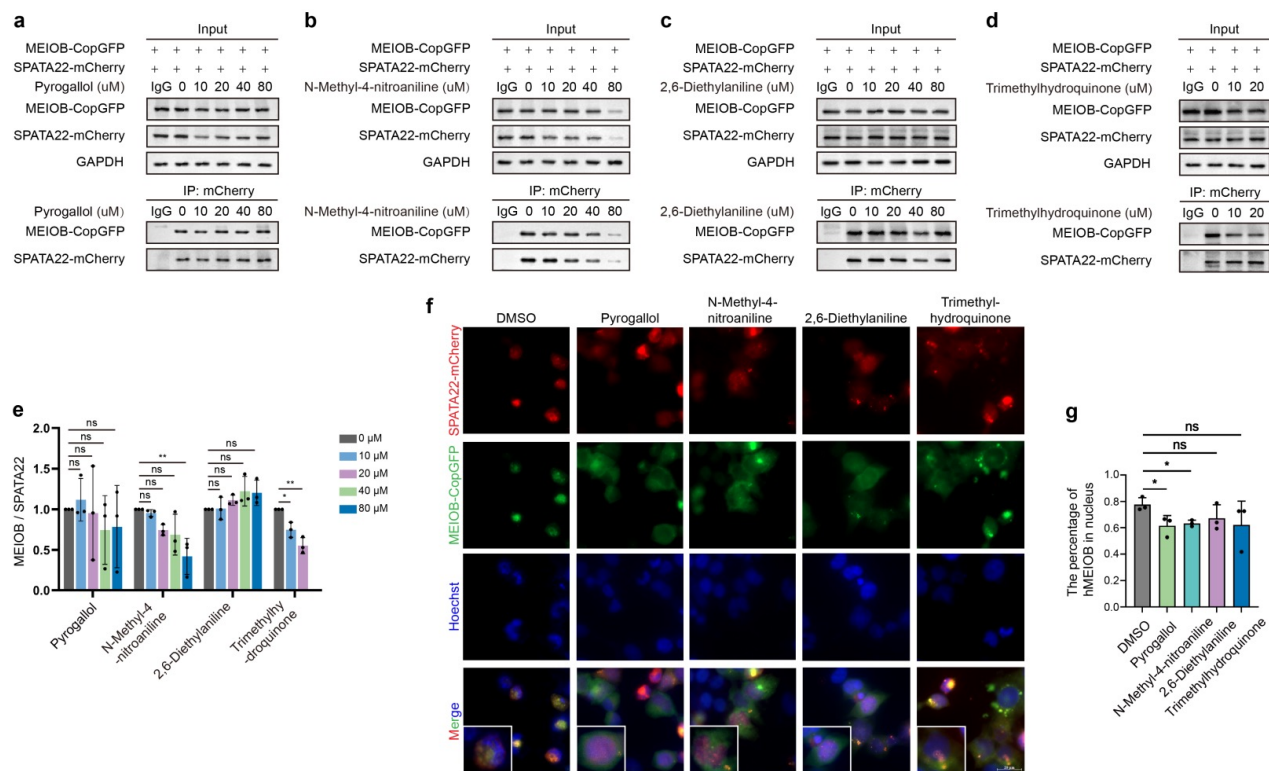


Figure 5 Effect of Small Molecules on the Interaction between MEIOB and SPATA22 and affect MEIOB nuclear localization. **a-d** MEIOB-CopGFP expressing plasmid was co-transfected with SPATA22-mCherry expressing plasmid into HEK 293T cell, and co-incubated with candidate molecules at different concentrations for 48 h. mCherry IP experiment was conducted on the cell lysates. IgG IP was set as a negative control. **e** Effects of candidate molecules at different concentrations on the interaction between MEIOB and SPATA22 in HEK 293T cells. Following IP of samples using the mCherry antibody, the ratio of the gray values of MEIOB protein to SPATA22 protein was determined, and normalization was performed using the ratio from the 0 μ M candidate molecule group as a reference. All results were obtained from three independent replicates. Data were analyzed using the two-way ANOVA and presented as mean $\pm$ SD (* P < 0.5, ** P < 0.01). **f** The distribution of MEIOB-CopGFP (green), SPATA22-mCherry (red), and Hoechst (blue) in HEK 293T cells. MEIOB-CopGFP expressing plasmid was co-transfected with SPATA22-mCherry expressing plasmid into HEK 293T cells, and co-incubated with candidate molecules at different concentrations for 48 h. Scale bar: 20 μ m. **g** Statistical analysis of the distribution of MEIOB-CopGFP in the cytoplasm and nucleus of HEK 293T cells co-expressing MEIOB-CopGFP and SPATA22-mCherry. The experiment was repeated three times, with the localization of 100 cells counted each time; data were analyzed using the unpaired t-test and presented as mean $\pm$ SD (* P < 0.5).

while ensuring that potentially effective molecules are not overlooked. To achieve this, we designed a hybrid deep learning model that integrates attention mechanisms with CNNs to perform the DTI task. To address the limitations of existing datasets, we further developed and implemented two distinct versions of the model, each optimized for specific tasks within the screening process.

Specifically, the first training strategy aims to eliminate as many ineffective cases as possible while retaining the majority of effective ones. To achieve this, we selected the top three negative samples with the highest fingerprint similarity from ChemDiv and ChemBridge out of the 177 effective cases, which were then used to construct the training and testing sets. The second training strategy focuses on maximizing the model's ability to recall as many true effective cases as possible. Considering that the binding affinities of effective cases with MEIOB-SPATA22 vary, we classified the effective cases based on their distribution, dividing them into 19 true effective cases and 158 false effective cases, with a threshold of -4.2 . In this case, the false-effective cases and the ineffective cases generated in the first stage were used to form the training and validation sets, while the true-effective cases were reserved for the testing set.

As illustrated in the first confusion matrix, the first training strategy achieves an accuracy score of 0.77, effectively eliminating a large proportion of ineffective cases while preserving a substantial fraction of positive candidates (Fig. 2a). Such a result demonstrates its robustness in distinguishing interacting from non-interacting compounds. Moreover, the second training strategy achieves a perfect recall score of 1.00 during the test phase (Fig. 2b). We present the prediction confidence level of all 19 test molecules (Fig. 2c). Not only are all 19 true effective cases sorted out, but 11 of them are also recalled with confidence greater than 0.6, exhibiting superior sensitivity.

Affinity prediction and ranking

Following the initial screening, the second stage of our solution focuses on predicting and ranking the binding affinities of the selected molecular candidates. This stage is crucial for refining the selection process and prioritizing compounds with the highest likelihood of effective interaction with MEIOB-SPATA22. Therefore, we proposed a dedicated DTA model, pre-trained under natural language supervision and fine-tuned by approaching regression values, to estimate the binding strengths, providing a quantitative assessment of molecular affinity.

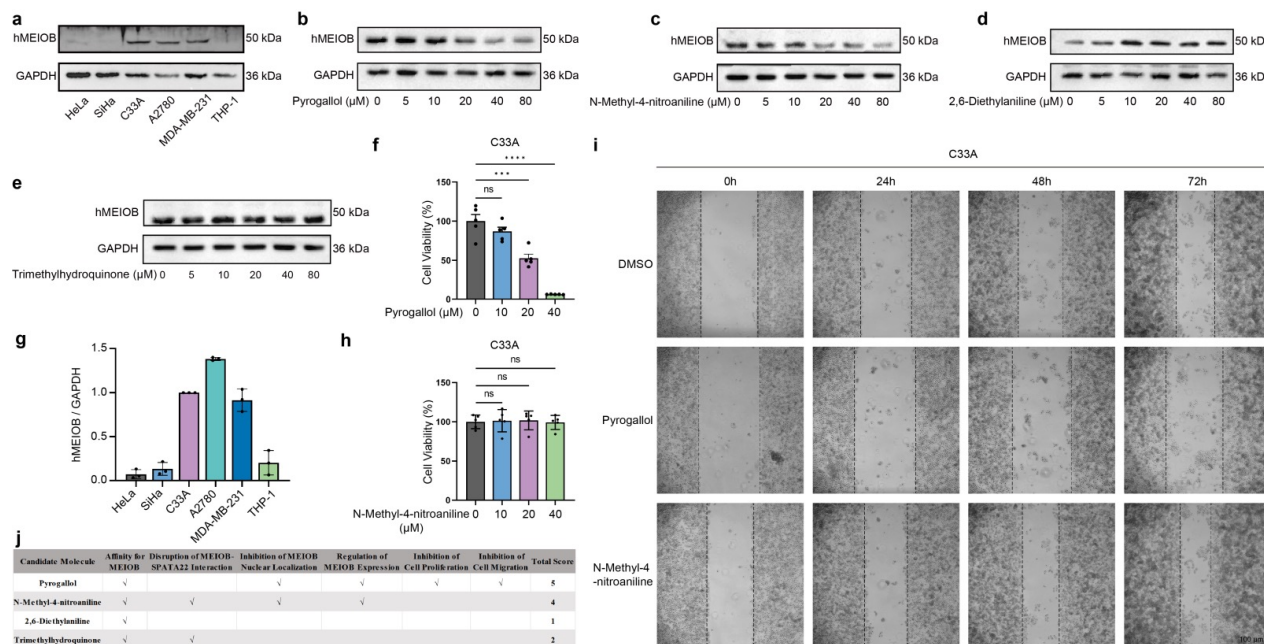


Figure 6 Small molecules inhibit MEIOB expression and cellular functions in C33A Cells. **a** Detect the expression of hMEIOB in various tumor cells (HeLa, SiHa, C33A, A2780, MDA-MB-231, THP-1). **b** The expression level of hMEIOB in C33A cells after treatment with pyrogallol for 48 h. **c** The expression level of hMEIOB in C33A cells after treatment with N-methyl-4-nitroaniline for 48 h. **d** The expression level of hMEIOB in C33A cells after treatment with 2,6-diethylaniline for 48 h. **e** The expression level of hMEIOB in C33A cells after treatment with trimethylhydroquinone for 48 h. **f** The cell viability of C33A cells after treatment with pyrogallol for 24 h. **g** Expression levels of hMEIOB protein in different tumor cell lines (HeLa, SiHa, C33A, A2780, MDA-MB-231, THP-1). Protein expression levels were quantified by gray values. The relative expression level of hMEIOB was normalized using GAPDH as an internal reference (expressed as the hMEIOB/GAPDH ratio), and then further normalized to the hMEIOB/GAPDH ratio of the C33A group in each experimental set. Band quantification was normalized to GAPDH. All experiments were independently repeated three times, and the quantitative results were presented as mean±SD. **h** The cell viability of C33A cells after treatment with N-methyl-4-nitroaniline for 24 h. **i** The wound-healing assay was employed to assess the impacts of pyrogallol and N-methyl-4-nitroaniline on the migratory capacity of C33A cells. **j** Scoring of MEIOB-related assays for candidate molecules. Each positive result counted as 1 point.

To evaluate our DTA model's performance, we integrated 177 effective cases with specific affinities into KIBA, and randomly split them into training and test sets. Notably, 29 effective cases were divided into the test set, and we provide a detailed visualization for testing analysis (Fig. 2d). The data distribution predicted by our DTA model is basically consistent with the true affinities of effective cases. Specifically, it achieves 0.834 in Rank score and 0.598 in PCC score, which reinforces the reliability of our model, as well as exhibits a more precise and biologically relevant ranking of candidate molecules. In practical applications, only a subset of top-ranked molecules is typically selected for experimental validation. A higher Rank score is particularly important in drug screening, as it reflects the model's ability to rank molecules based on predicted binding affinities.

Additionally, we conducted a comparative analysis of the DTA model of MMCDS against SOTA methods, including machine learning techniques such as random forest (RF) (Ballester and Mitchell 2010; Li et al. 2015), and deep learning models like MSGNN-DTA (Wang et al. 2023), HGRL-DTA (Chu et al. 2022), and MDCT-DTA (Zhu et al. 2024b).

Since our drug discovery process is a task for unseen drugs, which is an OOD problem, we designed the scenario where the drugs in the test set are absent from the training set for our experiment. In this way, we will effectively assess the OOD performance of different DTA

models. MMCDS outperforms all comparative baseline models across all criteria on both Davis and KIBA datasets (Fig. 2e and Fig. 2f). Specifically, MMCDS achieved improvements of 4.8%, 16.3%, and 9.6% in MSE, PCC, and SRCC, respectively, on the Davis dataset. Additionally, it demonstrated enhancements of 1.9%, 7.3%, and 2.2% in MSE, PCC, and SRCC, respectively, on the KIBA dataset, showcasing its superior generalization capability for OOD problems.

Large-scale virtual screening and candidate prioritization

Initially, by sequentially applying two different strategies to train the DTI models for filtering the initial pool of 163,132 small molecules, 2,324 molecular candidates demonstrating potential interactions with MEIOB-SPATA22 were successfully identified. Afterwards, our DTA model predicted the affinity scores of the 2,324 molecular candidates. We integrate predicted affinity scores with Schrödinger docking results to comprehensively evaluate each candidate. Based on this multifaceted assessment, these candidate molecules with the greatest potential were ultimately screened out. Finally, compounds with poor druggability were excluded, and 16 compounds that are currently readily accessible were selected for subsequent research. Our multi-stage, multi-model solution completes the stages of training, screening, prediction, and ranking within a few days, systematically evaluating and selecting potential drug candidates,

demonstrating its significant potential in accelerating the discovery of novel therapeutics with high specificity and efficacy.

SPATA22 functions as a chaperone protein for MEIOB. The absence of SPATA22 gives rise to the ablation of MEIOB functionality (Luo et al. 2013; Xu et al. 2017, 2022). Previously, we have successfully tested the binding affinity between proteins and small molecules using biolayer interferometry (BLI) technology (Xu et al. 2025). Therefore, to screen for small molecules that are capable of engaging in true competitive binding to MEIOB against SPATA22, we expressed full-length soluble MEIOB and used 8×His-Tag as a fusion protein, which was used to bind to NTA biosensors to assess the affinity of candidate molecules, specifically the KD value (Fig. 3a). The 16 screened candidate molecules were detected using BLI technology (Supplementary Table S1). After statistical analysis and removing the candidate molecules with overly fluctuating curves, the results showed that the KD values of 4 candidate molecules binding to MEIOB reached the 10^{-6} M, including pyrogallol, N-methyl-4-nitroaniline, trimethylhydroquinone, and 2,6-diethylaniline. Among them, the KD values of pyrogallol and N-methyl-4-nitroaniline reached 1.72×10^{-7} M and 2.06×10^{-7} M (Fig. 3b-e). These four candidate molecules specifically bind to MEIOB.

To investigate the action sites through which these candidate molecules interfere with the interaction between MEIOB and SPATA22, we predicted the structures of MEIOB (aa272-471) and SPATA22 (aa236-363) and constructed computational models of these candidate molecules binding to the complex. The model showed that pyrogallol inserts into the pocket of the MEIOB-SPATA22 complex and interacts with the SER 462 residue of MEIOB via hydrogen bonding; N-methyl-4-nitroaniline interacts with the SER 462, SER 466, and ARG 429 residues of MEIOB; 2,6-diethylaniline associates with the SER 462 and SER 466 residues of MEIOB; and trimethylhydroquinone acts on the SER 466 and ARG 429 sites of MEIOB (Supplementary Fig. S1). Collectively, BLI assays and molecular docking experiments confirmed that the four lead compounds exhibit high-affinity binding to MEIOB, with distinct binding sites mapped at key residues (e.g., SER 462, SER 466, ARG 429) within the MEIOB-SPATA22 interaction interface. These results not only validate the specific binding capacity of the identified compounds but also underscore the reliability and predictive power of MMCDS in screening high-potential lead molecules.

Biological Validation

We then performed biological validation to determine whether the computationally selected candidates exert measurable effects on MEIOB and the MEIOB–SPATA22 axis. This section includes validation of interaction disruption, assessment of MEIOB nuclear localization, evaluation of MEIOB expression in C33A cells, and examination of downstream cellular phenotypes such as proliferation and migration.

N-Methyl-4-nitroaniline and Trimethylhydroquinone Inhibits MEIOB-SPATA22 Interaction

MEIOB interacts with SPATA22, and the aa294-450 region of MEIOB serves as the binding domain for SPATA22. During meiosis, MEIOB can only be recruited to DNA double-strand breaks (DSBs) sites to exert its normal function in the presence of SPATA22 (Bearak et al. 2020; Ilani et al. 2012). To investigate whether the candidate molecules could disrupt the interaction between MEIOB

and SPATA22, we co-transfected CopGFP-tagged MEIOB and mCherry-tagged SPATA22 in cells and treated them with different concentrations of candidates (pyrogallol, N-methyl-4-nitroaniline, trimethylhydroquinone, and 2,6-diethylaniline) for detection. Results showed that after treatment with trimethylhydroquinone at concentrations of 10 μ M and 20 μ M (note: high concentrations of trimethylhydroquinone exhibited relatively strong cytotoxicity), the level of MEIOB bound to SPATA22 decreased with increasing drug concentration. Similar results were observed in cells treated with N-methyl-4-nitroaniline; however, no obvious effect was detected following treatment with pyrogallol or 2,6-diethylaniline (Fig. 4a-e). Collectively, these findings indicate that trimethylhydroquinone and N-methyl-4-nitroaniline can disrupt the interaction between MEIOB and SPATA22.

Pyrogallol and N-Methyl-4-nitroaniline Affect MEIOB Nuclear Localization

In the presence of SPATA22, MEIOB translocates into the nucleus, and the two proteins localize at DSBs sites to exert repair functions. When SPATA22 is absent in cells or the interaction between the two is disrupted, MEIOB cannot expose its nuclear localization signal (NLS) to enter the nucleus (Xu et al. 2022). To investigate whether candidates affect the nuclear localization of MEIOB, we co-transfected CopGFP-MEIOB and mCherry-SPATA22, and observed the localization of MEIOB and SPATA22 after candidate treatment. Imaging and statistical results showed that in the absence of candidates, MEIOB co-localized with SPATA22 in the nucleus. However, after treatment with pyrogallol and N-methyl-4-nitroaniline, the nuclear localization of MEIOB decreased, and it was instead expressed in the cytoplasm. However, no significant change in the nuclear localization of MEIOB was observed after treatment with 2,6-diethylaniline and trimethylhydroquinone (Fig. 5a,b). Collectively, cell fluorescence imaging and statistical analyses demonstrated that pyrogallol and N-methyl-4-nitroaniline specifically disrupt the nuclear localization of MEIOB, whereas 2,6-diethylaniline and trimethylhydroquinone exert no significant effect on the subcellular localization of MEIOB. These findings confirm that specific candidate molecules can interfere with the nuclear translocation of MEIOB, likely by disrupting the interaction between MEIOB and SPATA22.

Pyrogallol and N-Methyl-4-nitroaniline Inhibit the Expression of MEIOB in C33A Cells

Previous studies have found that MEIOB is involved in the repair of DSBs during meiotic homologous recombination. It not only participates in the regulation of spermatogenesis but also, as a cancer-testis gene, regulates the occurrence and development of diseases in tumors such as breast cancer, lung adenocarcinoma, and melanoma, showing a tumor-promoting effect (Gu et al. 2021; Saad and Hamed 2024; Shiohama et al. 2014). Therefore, we intend to test the potential of candidate molecules for cancer therapy in subsequent studies. First, we screened out several cell lines that may highly express MEIOB through the Human Protein Atlas (HPA) database (<https://www.proteinatlas.org/>) and confirmed by WB that MEIOB was highly expressed in the cervical cancer cell line C33A, ovarian cancer cell line A2780, and breast cancer cell line MDA-MB-231. Among these cell lines, A2780 exhibited the highest MEIOB expression level, followed by C33A and MDA-MB-231 (Fig. 5c, d). However, due to the relatively low incidence rate of ovarian cancer, we selected the

cervical cancer cell line C33A, which has high expression of MEIOB and a high incidence rate, for validation in subsequent experiments.

Next, we treated C33A cells with pyrogallol, N-methyl-4-nitroaniline, 2,6-diethylaniline, and trimethylhydroquinone. It was found that after treatment with pyrogallol or N-methyl-4-nitroaniline for 48 h, the protein level of MEIOB in the cells was down-regulated with the increase in the concentration of the small molecule, showing an inhibitory effect on MEIOB (Fig. 5e,f). 2,6-Diethylaniline and trimethylhydroquinone had no inhibitory effect on MEIOB (Fig. 6g,h).

Pyrogallol Inhibits the Proliferation and Migration Abilities of C33A Cells

Based on the aforementioned experimental data, pyrogallol and N-methyl-4-nitroaniline were shown to inhibit the expression of MEIOB in cervical cancer cells. To understand whether pyrogallol and N-methyl-4-nitroaniline affect the viability of cervical cancer cells, we treated C33A cells with different concentrations of candidate molecules and detected their viability at different times. The CCK-8 assay results showed that the proliferation capacity of C33A cells in the pyrogallol-treated group was significantly inhibited, and the cell survival rate decreased with the increase in the concentration of the candidate molecule (Fig. 6a). In contrast, no significant change in cell viability was observed in the N-methyl-4-nitroaniline-treated group (Fig. 6b).

Subsequently, we treated C33A cells with pyrogallol and N-methyl-4-nitroaniline and collected images at 0 h, 24 h, 48 h, and 72 h after treatment to observe the effect of candidate molecules on the migration ability of cervical cancer cells. It was found that after treatment with 80 μ M pyrogallol, the migration ability of C33A cells decreased significantly, whereas no obvious change was observed in the migration ability of cells in the N-methyl-4-nitroaniline-treated group (Fig. 6c). These results demonstrate that pyrogallol significantly inhibits the proliferation and migration capacities of C33A cells, and this effect is most likely mediated by downregulating MEIOB expression.

Next, we scored the candidate molecules, with the scoring criteria including affinity for MEIOB, regulation of MEIOB expression, disruption of MEIOB-SPATA22 interaction, regulation of MEIOB expression,, inhibition of cell proliferation, and inhibition of cell migration. Each positive test was counted as 1 point. The data showed that pyrogallol exerts the strongest inhibitory effect on MEIOB function, and this effect is likely mediated by disrupting the MEIOB-SPATA22 interaction and inhibiting MEIOB expression (Fig. 6d).

Conclusion

This study presents a multi-stage, multi-model hybrid screening solution for male contraceptive compound screening. By targeting two key proteins, MEIOB and SPATA22, our solution, MMCDS, facilitated the efficient identification of candidate compounds and screened out 4 compounds with micromolar potency. This innovative solution addresses long-standing issues in male contraceptive development, including low screening efficiency and extreme class imbalance (Ghosh et al. 2024; Rezvani and Wang 2023; Tarawneh et al. 2022). Compared with the current SOTA methods (Singh et al. 2022; Vidal-Limon et al. 2022), MMCDS achieves a 3.80-10.23% improvement in candidate compound screening accuracy.

Overall, this study provides a novel solution for AI-driven screening of male contraceptive compounds. In the face of challenges such as limited data and inconsistent labeling, MMCDS offers a generalizable paradigm that can be extended to small-molecule drug discovery in other therapeutic areas.

References

- P. J. Ballester and J. B. Mitchell. A machine learning approach to predicting protein–ligand binding affinity with applications to molecular docking. *Bioinformatics*, 26(9):1169–1175, 2010.
- J. Bearak, A. Popinchalk, B. Ganatra, A.-B. Moller, Ö. Tunçalp, C. Beavin, L. Kwok, and L. Alkema. Unintended pregnancy and abortion by income, region, and the legal status of abortion: estimates from a comprehensive model for 1990–2019. *The Lancet Global Health*, 8(9):e1152–e1161, 2020.
- Z. Chu, F. Huang, H. Fu, Y. Quan, X. Zhou, S. Liu, and W. Zhang. Hierarchical graph representation learning for the prediction of drug-target binding affinity. *Information Sciences*, 613:507–523, 2022.
- M. I. Davis, J. P. Hunt, S. Herrgard, P. Ciceri, L. M. Wodicka, G. Pallares, M. Hocker, D. K. Treiber, and P. P. Zarrinkar. Comprehensive analysis of kinase inhibitor selectivity. *Nature biotechnology*, 29(11):1046–1051, 2011.
- Y. Feng, Y. Long, H. Wang, Y. Ouyang, Q. Li, M. Wu, and J. Zheng. Benchmarking machine learning methods for synthetic lethality prediction in cancer. *Nature Communications*, 15(1):9058, 2024.
- M. Gershoni, R. Hauser, L. Yogev, O. Lehavi, F. Azem, H. Yavetz, S. Pietrokovski, and S. E. Kleiman. A familial study of azoospermic men identifies three novel causative mutations in three new human azoospermia genes. *Genetics in Medicine*, 19(9):998–1006, 2017.
- K. Ghosh, C. Bellingier, R. Corizzo, P. Branco, B. Krawczyk, and N. Japkowicz. The class imbalance problem in deep learning. *Machine Learning*, 113(7):4845–4901, 2024.
- Y. Gu, C. Wang, R. Zhu, J. Yang, W. Yuan, Y. Zhu, Y. Zhou, N. Qin, H. Shen, H. Ma, et al. The cancer-testis gene, meioB, sensitizes triple-negative breast cancer to parp1 inhibitors by inducing homologous recombination deficiency. *Cancer biology & medicine*, 18(1):74–87, 2021.
- R. Guo, Y. Xu, N. A. Leu, L. Zhang, S. Y. Fuchs, L. Ye, and P. J. Wang. The ssdna-binding protein meioB acts as a dosage-sensitive regulator of meiotic recombination. *Nucleic acids research*, 48(21):12219–12233, 2020.
- T. He, M. Heidemeyer, F. Ban, A. Cherkasov, and M. Ester. Simboost: a read-across approach for predicting drug–target binding affinities using gradient boosting machines. *Journal of cheminformatics*, 9:1–14, 2017.
- K. E. Hevener, R. Pesavento, J. Ren, H. Lee, K. Ratia, and M. E. Johnson. Hit-to-lead: hit validation and assessment. In *Methods in Enzymology*, volume 610, pages 265–309. Elsevier, 2018.
- S.-H. Hong, G. Castro, D. Wang, R. Nofsinger, M. Kane, A. Folias, A. R. Atkins, R. T. Yu, J. L. Napoli, P. Sassone-Corsi, et al. Targeting nuclear receptor corepressors for reversible male contraception. *Proceedings of the National Academy of Sciences*, 121(9):e2320129121, 2024.
- N. Ilani, M. Y. Roth, J. K. Amory, R. S. Swerdloff, C. Dart, S. T. Page, W. J. Bremner, R. Sitruk-Ware, N. Kumar, D. L. Blithe, et al. A new combination of testosterone and nesterone

- transdermal gels for male hormonal contraception. *The Journal of Clinical Endocrinology & Metabolism*, 97(10):3476–3486, 2012.
- S. La Salle, K. Palmer, M. O'Brien, J. C. Schimenti, J. Eppig, and M. A. Handel. Spata22, a novel vertebrate-specific gene, is required for meiotic progress in mouse germ cells. *Biology of reproduction*, 86(2):45–1, 2012.
- H. Li, K.-S. Leung, M.-H. Wong, and P. J. Ballester. Low-quality structural and interaction data improves binding affinity prediction via random forest. *Molecules*, 20(6):10947–10962, 2015.
- K. Li, X. Gong, J. Wu, and W. Hu. Contrastive learning drug response models from natural language supervision. In K. Larson, editor, *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, pages 2126–2134. International Joint Conferences on Artificial Intelligence Organization, 8 2024a. doi: 10.24963/ijcai.2024/235. URL <https://doi.org/10.24963/ijcai.2024/235>. Main Track.
- K. Li, W. Liu, Y. Luo, X. Cai, J. Wu, and W. Hu. Zero-shot learning for preclinical drug screening. In K. Larson, editor, *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*, pages 2117–2125. International Joint Conferences on Artificial Intelligence Organization, 8 2024b. doi: 10.24963/ijcai.2024/234. URL <https://doi.org/10.24963/ijcai.2024/234>. Main Track.
- K. Li, Y. Xiong, H. Zhang, X. Cai, J. Wu, B. Du, and W. Hu. Graph-structured small molecule drug discovery through deep learning: Progress, challenges, and opportunities. In *2025 IEEE International Conference on Web Services (ICWS)*, pages 1033–1042, 2025a.
- K. Li, Y. Zeng, Y.-d. Xiong, H.-c. Wu, S. Fang, Z.-y. Qu, Y. Zhu, B. Du, Z.-b. Gao, and W.-b. Hu. Contrastive learning-based drug screening model for glun1/glun3a inhibitors. *Acta Pharmacologica Sinica*, Jun 2025b. ISSN 1745-7254. doi: 10.1038/s41401-025-01580-0. URL <https://doi.org/10.1038/s41401-025-01580-0>.
- Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou. A survey of convolutional neural networks: analysis, applications, and prospects. *IEEE transactions on neural networks and learning systems*, 33(12):6999–7019, 2021.
- L. Liu, Q. Dou, H. Chen, J. Qin, and P.-A. Heng. Multi-task deep model with margin ranking loss for lung nodule analysis. *IEEE transactions on medical imaging*, 39(3):718–728, 2019.
- Y. Liu, L. Xing, L. Zhang, H. Cai, and M. Guo. Geformerdt: drug target affinity prediction based on transformer graph for early fusion. *Scientific Reports*, 14(1):7416, Mar 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-57879-1. URL <https://doi.org/10.1038/s41598-024-57879-1>.
- M. Luo, F. Yang, N. A. Leu, J. Landaiche, M. A. Handel, R. Benavente, S. La Salle, and P. J. Wang. Meiob exhibits single-stranded dna-binding and exonuclease activities and is essential for meiotic recombination. *Nature communications*, 4(1):2788, 2013.
- N. Mannowetz, S. S. Chung, S. Maitra, M. A. A. Noman, H. L. Wong, N. Cheryala, A. Bakshi, D. J. Wolgemuth, and G. I. Georg. Targeting the retinoid signaling pathway with yct-529 for effective and reversible oral contraception in mice and primates. *Communications Medicine*, 5(1):68, 2025.
- K. Mavrogiorgos, A. Kiourtis, A. Mavrogiorgou, A. Menychtas, and D. Kyriazis. Bias in machine learning: A literature review. *Applied Sciences*, 14(19):8860, 2024.
- S. Moon, W. Zhung, S. Yang, J. Lim, and W. Y. Kim. Pignet: a physics-informed deep learning model toward generalized drug-target interaction predictions. *Chemical Science*, 13(13):3661–3673, 2022.
- F. Noor, M. Junaid, A. H. Almalki, M. Almaghrabi, S. Ghazanfar, and M. Tahir ul Qamar. Deep learning pipeline for accelerating virtual screening in drug discovery. *Scientific Reports*, 14(1):28321, 2024.
- M. Pandey, M. Fernandez, F. Gentile, O. Isayev, A. Tropsha, A. C. Stern, and A. Cherkasov. The transformational role of gpu computing and deep learning in drug discovery. *Nature Machine Intelligence*, 4(3):211–221, 2022.
- S. Rezvani and X. Wang. A broad review on class imbalance learning techniques. *Applied Soft Computing*, 143:110415, 2023.
- J. Ribeiro, P. Dupaigne, C. Petrillo, C. Ducrot, C. Duquenne, X. Veaute, C. Saintomé, D. Busso, R. Guerois, E. Martini, et al. The meiosis-specific meiob-spata22 complex cooperates with rpa to form a compacted mixed meiob/spata22/rpa/ssdna complex. *Dna Repair*, 102:103097, 2021.
- M. N. Saad and M. Hamed. Transcriptome-wide association study reveals new molecular interactions associated with melanoma pathogenesis. *Cancers*, 16(14):2517, 2024.
- F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini. The graph neural network model. *IEEE transactions on neural networks*, 20(1):61–80, 2008.
- S. Sharma, D. Ahire, A. Basit, M. Lajoie, C. Wang, M. S. Lee, D. L. Blithe, J. K. Amory, D. K. Singh, S. Heyward, et al. Dimethandrolone, a potential male contraceptive pill, is primarily metabolized by the highly polymorphic udp-glucuronosyltransferase 2b17 enzyme in human intestine and liver. *Drug Metabolism and Disposition*, 50(12):1493–1500, 2022.
- Y. Shiohama, J. Ohtake, T. Ohkuri, D. Noguchi, Y. Togashi, H. Kitamura, and T. Nishimura. Identification of a meiosis-specific protein, meiob, as a novel cancer/testis antigen and its augmented expression in demethylated cancer cells. *Immunology letters*, 158(1-2):175–182, 2014.
- S. Singh, Q. B. Baker, and D. B. Singh. Molecular docking and molecular dynamics simulation. In *Bioinformatics*, pages 291–304. Elsevier, 2022.
- B. Souquet, E. Abby, R. Hervé, F. Finsterbusch, S. Tourpin, R. Le Bouffant, C. Duquenne, S. Messiaen, E. Martini, J. Bernardino-Sgherri, et al. Meiob targets single-strand dna and is necessary for meiotic recombination. *PLoS genetics*, 9(9):e1003784, 2013.
- A. S. Tarawneh, A. B. Hassanat, G. A. Altarawneh, and A. Almuhaimeed. Stop oversampling for class imbalance learning: A review. *IEEE Access*, 10:47643–47660, 2022.
- C. Tian, L. Wang, Z. Cui, and H. Wu. Gtamp-dta: Graph transformer combined with attention mechanism for drug-target binding affinity prediction. *Computational Biology and Chemistry*, 108:107982, 2024. ISSN 1476-9271. doi: <https://doi.org/10.1016/j.compbiolchem.2023.107982>. URL <https://www.sciencedirect.com/science/article/pii/S1476927123001731>.
- P. C. Tiwari, R. Pal, M. J. Chaudhary, and R. Nath. Artificial intelligence revolutionizing drug development: Exploring opportunities and challenges. *Drug Development Research*, 84(8):1652–1663, 2023.

- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin. Attention is all you need. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.
- A. Vidal-Limon, J. E. Aguilar-Toalá, and A. M. Liceaga. Integration of molecular docking analysis and molecular dynamics simulations for studying food proteins and bioactive peptides. *Journal of agricultural and food chemistry*, 70(4): 934–943, 2022.
- I. Wallach, D. Bernard, K. Nguyen, G. Ho, A. Morrison, A. Stecula, A. Rosnik, A. M. O’Sullivan, A. Davtyan, B. Samudio, et al. Ai is a viable alternative to high throughput screening: a 318-target study. *Scientific Reports*, 14(1):7526, 2024.
- S. Wang, X. Song, Y. Zhang, K. Zhang, Y. Liu, C. Ren, and S. Pang. Msgnn-dta: Multi-scale topological feature fusion based on graph neural networks for drug–target binding affinity prediction. *International Journal of Molecular Sciences*, 24(9):8326, 2023.
- Y. Wu, Y. Li, G. Murtaza, J. Zhou, Y. Jiao, C. Gong, C. Hu, Q. Han, H. Zhang, Y. Zhang, et al. Whole-exome sequencing of consanguineous families with infertile men and women identifies homologous mutations in spata22 and meiob. *Human Reproduction*, 36(10):2793–2804, 2021.
- Y. Xu, R. A. Greenberg, E. Schonbrunn, and P. J. Wang. Meiosis-specific proteins meiob and spata22 cooperatively associate with the single-stranded dna-binding replication protein a complex and dna double-strand breaks. *Biology of reproduction*, 96(5):1096–1104, 2017.
- Y. Xu, R. Liu, N. A. Leu, L. Zhang, I. Ibragmova, D. C. Schultz, and P. J. Wang. A cell-based high-content screen identifies isocotoin as a small molecule inhibitor of the meiosis-specific meiob–spata22 complex. *Biology of reproduction*, 103(2):333–342, 2020.
- Y. Xu, Z. Chen, P. Wu, W. Qu, H. Shi, M. Cheng, Y. Xu, T. Jin, C. Liu, Y. Li, et al. Nuclear localization of human meiob requires its nls in the ob domain and interaction with spata22: Human spata22 assists meiob to enter the nucleus. *Acta Biochimica et Biophysica Sinica*, 55(1):154, 2022.
- Y. Xu, W. Qu, E. Zhou, Q. Sun, W. Gong, L. Xu, Y. Lei, Z. Jia, H. Shi, X. Zhang, et al. New feature of hmeiob and hspata22 binding to ssdna from a single-molecule perspective. *Acta biochimica et biophysica Sinica*, 2025.
- J. Yang, K. Zhou, Y. Li, and Z. Liu. Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132(12):5635–5662, 2024.
- Z. Yang, Y. Yu, C. You, J. Steinhardt, and Y. Ma. Rethinking bias-variance trade-off for generalization of neural networks. In *International Conference on Machine Learning*, pages 10767–10777. PMLR, 2020.
- C. Yao, D. Hou, Z. Ji, D. Pang, P. Li, R. Tian, Y. Zhang, N. Ou, H. Bai, E. Zhi, et al. Bi-allelic spata22 variants cause premature ovarian insufficiency and nonobstructive azoospermia due to meiotic arrest. *Clinical Genetics*, 101(5-6):507–516, 2022.
- M. Yazdani-Jahromi, N. Yousefi, A. Tayebi, E. Kolanthai, C. J. Neal, S. Seal, and O. O. Garibay. Attentionsitedti: an interpretable graph-based model for drug-target interaction prediction using nlp sentence-level relation classification. *Briefings in Bioinformatics*, 23(4):bbac272, 2022.
- Q. Ye and Y. Sun. Graph neural pre-training based drug-target affinity prediction. *Frontiers in Genetics*, 15:1452339, 2024.
- Q. Ye, C.-Y. Hsieh, Z. Yang, Y. Kang, J. Chen, D. Cao, S. He, and T. Hou. A unified drug–target interaction prediction framework based on knowledge graph and recommendation system. *Nature communications*, 12(1):6775, 2021.
- Y. Zang, M. Su, Q. Wang, X. Cheng, W. Zhang, Y. Zhao, T. Chen, Y. Jiang, Q. Shen, J. Du, et al. High-throughput screening of sars-cov-2 main and papain-like protease inhibitors. *Protein & Cell*, 14(1):17–27, 2023.
- X. Zhu, K. Hu, H. Cheng, H. Wu, K. Li, Y. Gao, M. Lv, C. Xu, H. Geng, Q. Shen, et al. Novel meiob pathogenic variants including a homozygous non-canonical splicing variant, cause meiotic arrest and human non-obstructive azoospermia. *Clinical Genetics*, 105(1):99–105, 2024a.
- Z. Zhu, X. Zheng, G. Qi, Y. Gong, Y. Li, N. Mazur, B. Cong, and X. Gao. Drug-target binding affinity prediction model based on multi-scale diffusion and interactive learning. *Expert Systems with Applications*, 255:124647, 2024b. ISSN 0957-4174. doi: <https://doi.org/10.1016/j.eswa.2024.124647>. URL <https://www.sciencedirect.com/science/article/pii/S0957417424015148>.